CMSC433 / Part 06

Verification conditions

Topic mock

Topic mock, using the sample's question styles. Attempt independently; feedback after the whole file. Untimed practice, no points assigned. Current exam scope and rules are unconfirmed.

Question. For each proposed invariant, list the induced initialization, preservation, and exit verification conditions for this method. Write the substituted formulas even when the invariant is invalid. Include the post-loop assignment. This task asks for partial correctness, not termination.

```
method ThreeTimes(n:int) returns (t:int)
  requires n > 0
  ensures t == 3*n + 2
{
  t := 0;
  var i := n;
  while i > 0 {
    t := t + 3;
    i := i - 1;
  }
  t := t + 2;
}
```

(a) $i \geq 0 \wedge t + 3i = 3n$.

Answer: Initialization: [ ].

Answer: Preservation: [ ].

Answer: Exit: [ ].

(b) $i \geq 0 \wedge t = 3i$.

Answer: Initialization: [ ].

Answer: Preservation: [ ].

Answer: Exit: [ ].

(c) $i \geq 0$.

Answer: Initialization: [ ].

Answer: Preservation: [ ].

Answer: Exit: [ ].